Alfred AI — Complete Developer Blueprint

The definitive architecture, build order, and technical specification

GoCodeMe • February 28, 2026

The Architecture Decision

This is not a list of options. This is the decision. Everything in this document follows from it.

Together.ai is Alfred's AI engine — his brain, voice, ears, and eyes.
The GoCodeMe server is Alfred's hands — he deploys, edits, manages, and executes.

Together.ai handles every AI inference call: speech to text, text to speech, language understanding, image generation, video generation, vision, and semantic search. One API. One key. Already configured. Already tested live on the server. 253 models available instantly with no GPU management, no model downloads, no local inference infrastructure.

The server handles everything that requires direct access to the hosting environment: deploying code, managing DNS, creating databases, running terminal commands, processing files, editing video, converting images, and downloading media. These are execution tasks, not AI inference tasks. They stay local because they have to — they touch the live hosting account directly.

This separation is clean, intentional, and permanent. When a developer sees a task in this document, the answer to "where does this run?" is always one of two things:

- **AI inference** !’ Together.ai API
- **Execution and file operations** !’ GoCodeMe server

What Alfred Is

Alfred is not a chatbot. Alfred is not a code editor plugin. Alfred is the world's first AI agent that manages your entire online presence, talks to you live like a person, builds and deploys your code, handles your hosting, generates media, and gets smarter with every interaction — all from one interface.

Cursor is a code editor trying to be an agent. It cannot deploy your DNS, create your database, send your email, or scan your server.

ChatGPT and Gemini are voice assistants with no ability to act on anything. They can talk but cannot do.

Alfred acts, speaks, listens, and builds. The combination of Together.ai's AI engine and GoCodeMe's hosting execution layer creates something that does not exist anywhere else.

The Business Case

A New Product Category

After this build GoCodeMe is no longer a hosting company with an AI assistant. It becomes the only product in the world where you talk to your AI, it talks back, builds your site, deploys it live, and manages everything — all in one place.

No one has this. Not GoDaddy. Not Hostinger. Not WP Engine. Not Cursor. Not ChatGPT. Not Gemini.

Revenue Impact

Retention: When Alfred knows a customer's entire site, speaks to them daily, and manages their infrastructure — leaving feels like firing a person, not cancelling a subscription. Churn drops fundamentally.

Upsell without selling: Alfred notices mid-conversation that a customer has no backup, no SSL, and is on a basic plan. He says "I noticed your site has no backup — want me to set one up right now?" That is an upsell that feels like help. Conversion will be multiples higher than any email campaign.

Premium pricing justified: Hosting is a commodity at $5 to $10 per month. An AI that manages your entire online presence and talks to you like a person is a $50 to $200 per month product. The comparison stops being other hosting companies and starts being a developer plus a sysadmin plus a designer on retainer.

Customer acquisition via demo: A 60-second demo — person talks to Alfred, Alfred talks back, builds a page, deploys it live, says "done, want to see it?" — will go viral. People share it because they have never seen anything like it.

French market advantage: Together.ai Whisper and XTTS v2 both support French natively. GoCodeMe targets Quebec and France. A French-speaking Alfred in a market where all AI tools are English-first is a dominant local advantage before any global competitor catches up.

White label and reseller: Once Alfred has voice plus full hosting management, agencies pay GoCodeMe to power their own AI assistant — their brand, their voice, their clients. A B2B revenue stream that does not exist today.

Together.ai — Alfred's AI Engine

The Together.ai API key is already configured on the server. All speed numbers below are real tests run live on this server before this document was written.

The Core Principle

Every AI inference task goes through Together.ai. No exceptions. This means:

- No local model management
- No GPU driver configuration
- No pip installs for AI models
- No 20-second model load times
- No RAM pressure from running local inference
- No maintenance when models are updated
- Best models in the world, available instantly

Speech to Text

Model	Tested Speed	Pricing	Detail
openai/whisper-large-v3	**1.13 seconds**	$0.27 per hour of audio	Transcribed 12.6 seconds of audio in 1.13s. Best accuracy available anywhere.

Text to Speech

Model	Tested Speed	Best Use	Pricing
hexgrad/Kokoro-82M	**0.73 seconds**	Alfred live chat voice — real-time conversation	$0.27 per million characters
cartesia/sonic-3	**3.93 seconds**	High quality, richer voice for longer responses	$65 per million characters
canopylabs/orpheus-3b-0.1-ft	**8.8 seconds**	Most expressive and emotional — voiceover generation	$0.27 per million characters
cartesia/sonic-2	Available	Alternative Cartesia voice	$65 per million characters
minimax/speech-2.6-turbo	Available	Additional TTS option	Available

The Live Voice Round Trip

```
User speaks !' Whisper large-v3 (1.13s) !' Alfred AI !' Kokoro 82M (0.73s) !' User hears Alfred
Total latency: approximately 2 seconds
```

This matches ChatGPT Advanced Voice Mode speed. No local GPU. No model downloads. Together.ai API.

AI Brain — Recommended Models by Task

Task	Model	Context	Pricing per 1M tokens
Read entire codebase	meta-llama/Llama-4-Scout-17B-16E-Instruct	**1,048,576 tokens**	$0.18 in / $0.59 out
Large context fast	meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8	**1,048,576 tokens**	$0.27 in / $0.85 out
General coding	deepseek-ai/DeepSeek-V3.1	131,072 tokens	$0.60 in / $1.70 out
Coding specialist	Qwen/Qwen3-Coder-480B-A35B-Instruct-FP8	262,144 tokens	$2.00 in / $2.00 out
Deep reasoning	deepseek-ai/DeepSeek-R1	163,840 tokens	$3.00 in / $7.00 out
Fast everyday chat	meta-llama/Llama-3.3-70B-Instruct-Turbo	131,072 tokens	$0.88 in / $0.88 out
Vision — read screenshots	Qwen/Qwen3-VL-32B-Instruct	262,144 tokens	$0.50 in / $1.50 out
Extended thinking	moonshotai/Kimi-K2-Thinking	262,144 tokens	$1.20 in / $4.00 out

Embeddings — Semantic Search

Model	Pricing	Use
intfloat/multilingual-e5-large-instruct	$0.02 per 1M	Semantic codebase search, any language
BAAI/bge-large-en-v1.5	$0.016 per 1M	English semantic search, fastest
mixedbread-ai/Mxbai-Rerank-Large-V2	$0.10 per 1M	Rerank results for accuracy

This solves the number one gap versus Cursor — semantic codebase understanding — with zero local infrastructure.

Image Generation — 27 Models Available

Recommended defaults:

Model	Best For
black-forest-labs/FLUX.2-pro	Highest quality — default for premium generation
google/imagen-4.0-ultra	Google's best — photorealistic
ideogram/ideogram-3.0	Text inside images — logos, banners, signs
black-forest-labs/FLUX.1-kontext-pro	Edit an existing image with a text prompt
black-forest-labs/FLUX.1-schnell	Fastest generation — previews and drafts

All 27 image models available on Together.ai including FLUX.1.1-pro, FLUX.2-flex, FLUX.2-dev, FLUX.1-kontext-max, Google Imagen 4 preview and fast, Google Flash Image 2.5, Google Gemini 3 Pro Image, Ideogram 3.0, ByteDance Seedream 4.0 and 3.0, HiDream I1 Full, Fast and Dev, Wan 2.6, Juggernaut Lightning Flux, Juggernaut Pro Flux, DreamShaper, Stable Diffusion XL, Stable Diffusion 3.

Video Generation — 23 Models Available

No other coding tool or AI IDE has this capability. Alfred generates video from text or from an image.

Recommended defaults:

Model	Type	Quality
google/veo-3.0-audio	Text to video with audio	Google flagship — best quality
openai/sora-2-pro	Text to video	OpenAI Sora Pro
kwaivgI/kling-2.1-master	Text to video	Cinematic quality
Wan-AI/Wan2.2-I2V-A14B	Image to video	Animate an existing image
ByteDance/Seedance-1.0-pro	Text to video	Fast, high quality

All 23 video models available including Google Veo 3.0, Veo 3.0 fast, Veo 3.0 fast audio, Veo 2.0, OpenAI Sora 2, Sora 2 Pro, Kling 2.1 master, pro and standard, Kling 2.0 master, Kling 1.6 pro and standard, ByteDance Seedance 1.0 pro and lite, Minimax Hailuo 02, Minimax Video 01 Director, Wan 2.2 T2V and I2V, PixVerse v5 and v5.6, Vidu Q1 and 2.0.

The Server — Alfred's Hands

The server handles everything that requires direct access to the hosting environment. These are not AI tasks — they are execution tasks. They stay local.

Hardware

Resource	Specification
CPU	Intel Xeon E-2386G @ 3.50GHz — 12 cores
RAM	31 GB total — 23 GB available
Disk	3.6 TB total — 3.2 TB free
OS	Ubuntu 22.04 LTS
CUDA	PyTorch built with CUDA 12.8 — GPU exposure needed to activate for local inference

What the Server Executes — Alfred's 87 Live Tools

Alfred already beats every competitor on hosting execution. None of the following exist in Cursor, ChatGPT, or Gemini:

Deploy code live to hosting, full DNS management, SSL certificate provisioning and HTTPS enforcement, domain registration and management, email account creation and management, email forwarding and autoresponders, send email from customer domains, WordPress installation and full WP-CLI management, MySQL database creation and management, malware scanning and security auditing, Word and PDF document generation, full billing and invoice management, shell command execution on the live server via run_terminal_command, Git checkpoints and restore, backup creation and restore, cron job management, traffic analytics and error log analysis, subdomain management, PHP version awareness.

Server-Side Tools for File and Media Processing

These run locally because they process files directly on the hosting account. Together.ai does not handle file manipulation — only AI inference.

Tool	Version	Handles
FFmpeg	7.0.2 static	Video and audio processing, conversion, streaming, trimming
MoviePy	2.1.2	Programmatic video editing in Python
ImageMagick	7.1.1	Advanced image processing, batch conversion, watermarking
Pillow	11.3.0	Python image manipulation
vtracer	0.6.12	Convert raster images to SVG
Librosa	0.11.0	Audio analysis and feature extraction
Basic Pitch	Installed	Convert audio files to MIDI
yt-dlp	2026.02.04	Download media from 1,864 sites
wkhtmltopdf	0.12.6	HTML to PDF conversion
mysql CLI	10.6.25	Direct database query execution

Runtimes Already on Server

PHP 8.3.30 and 8.2 available, Node.js 20.20.0, Python 3.10.12, Composer 2.9.5, WP-CLI 2.12.0, Git 2.34.1, npm 10.8.2, curl 7.81.0, wget 1.21.2, rsync 3.2.7, OpenSSL 3.0.2, Flask 3.1.2, aiohttp 3.13.3, requests 2.32.5.

XTTS v2 — Local Voice Fallback

The XTTS v2 model is fully downloaded and ready on the server. It is the fallback when Together.ai TTS is not needed — specifically for voice cloning, which requires a local model.

- Location: ~/.local/share/tts/tts_models--multilingual--multi-dataset--xtts_v2/
- Files present: config.json, model.pth, speakers_xtts.pth, vocab.json
- Built-in voices: 58 speakers — male and female
- Languages: English, Spanish, French, German, Italian, Portuguese, Polish, Turkish, Russian, Dutch, Czech, Arabic, Chinese, Hungarian, Korean, Japanese, Hindi
- Voice cloning: 6-second audio sample !’ cloned voice, any person
- Speed on CPU: 7 to 8 seconds
- Speed with GPU exposed: 0.3 to 0.5 seconds

Primary use: voice cloning for customer sites and agency white label. Together.ai Kokoro handles live chat voice.

What Needs to Be Installed

Four Pip Installs — That Is All

```
pip3 install websockets flask-socketio webrtcvad beautifulsoup4
```

Package	Purpose
websockets	Real-time bidirectional audio streaming between browser and server
flask-socketio	WebSocket server layer for the voice pipeline
webrtcvad	Voice Activity Detection — knows when the user has stopped speaking
beautifulsoup4	HTML parsing for the fetch_url tool

One System Package — Needs Server Admin

```
apt-get install espeak-ng
```

Unlocks local VITS fast TTS model as an additional fallback. Requires root access.

Nothing Else

All AI models are on Together.ai. All media processing tools are already on the server. All runtimes are already installed. The API key is already configured.

The Build Order

1. Live Two-Way Voice

Effort: 3 to 5 days

The highest impact feature. Nothing like this exists in any competing product.

Full specification in the next section.

2. Semantic Codebase Search

Effort: 2 days

Index public_html using Together.ai embeddings. Alfred answers "find everywhere we handle payments" semantically across every file. Beats Cursor @codebase directly.

- STT: intfloat/multilingual-e5-large-instruct for indexing
- Reranking: mixedbread-ai/Mxbai-Rerank-Large-V2 for result accuracy
- Storage: Simple JSON index on server — no vector database needed

3. Vision Input — Screenshot to Code

Effort: 2 days

User uploads or pastes a screenshot. Alfred reads it, understands what is broken or needed, writes the fix, deploys it live. Cursor reads screenshots. Alfred reads screenshots AND deploys.

- Model: Qwen/Qwen3-VL-32B-Instruct on Together.ai

4. Agentic Auto-Loop

Effort: 2 days

Alfred runs a terminal command, reads the output, fixes code if there is an error, re-runs, confirms success — without user prompting. The loop runs until the task succeeds.

- run_terminal_command is already live on the server
- Loop logic needs enabling in the agent runner

5. fetch_url Tool

Effort: 1 day

Alfred fetches any URL, strips it to readable text, uses it as context. "Build me a Stripe payment form using this API documentation: [url]" — Alfred reads the docs and builds it.

- Uses: curl already on server plus beautifulsoup4 from the four-package install
- Beats Cursor @web and @docs

6. Upgrade Image Generation

Effort: 1 day

Replace current single image model with the full Together.ai image library. Model selector in the tool. FLUX.1-kontext for editing existing images. Ideogram 3.0 for logos and text-in-image. Users pick the right model for the job.

7. generate_video Tool

Effort: 1 day

Alfred generates video from a text description or an existing image. "Create a 10-second promo video for my bakery" — Alfred generates it, saves it to the site, returns the URL.

- Default model: google/veo-3.0-audio
- Alternative: openai/sora-2, kwaivgI/kling-2.1-master

8. generate_audio Tool

Effort: 1 day

Alfred generates voiceovers, podcast intros, and audio content. "Read my about page as a natural voiceover" — Alfred generates an MP3 and delivers the URL.

- Expressive voice: canopylabs/orpheus-3b-0.1-ft
- Fast generation: hexgrad/Kokoro-82M

9. execute_sql Tool

Effort: half day

Alfred runs SQL against any customer database from chat. "Show me all orders over $500 this month" or "Fix the duplicate records in the users table."

- Uses: mysql CLI already on server
- Needs: safe parameter handling, read/write mode toggle

10. process_video Tool

Effort: 1 day

Alfred edits existing video — trim, merge, convert format, extract audio, add subtitles, resize for social media.

- Uses: FFmpeg 7.0.2 and MoviePy 2.1.2 already on server

11. process_image Tool

Effort: half day

Alfred manipulates existing images — resize, compress, convert format, watermark, batch process, convert raster to SVG.

- Uses: ImageMagick 7.1.1, Pillow 11.3.0, and vtracer already on server

12. download_media Tool

Effort: half day

Alfred downloads video or audio from any of 1,864 supported sites. "Download that YouTube tutorial and save it to my site."

- Uses: yt-dlp already on server

13. switch_php_version Tool

Effort: half day

Alfred switches a domain between PHP 8.2 and 8.3.

- Both binaries already at /usr/local/php82/bin/php and /usr/local/php83/bin/php

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Live Two-Way Voice — Full Specification

What It Is

The user speaks to Alfred. Alfred listens, understands, and responds in real time. Natural conversation. Like calling a person who manages your entire online presence and can fix your code, deploy your site, and manage your hosting while you are talking.

Exactly like ChatGPT Advanced Voice Mode and Google Gemini Live — except Alfred can act on everything he hears. ChatGPT and Gemini can talk. Alfred can talk AND do.

Architecture

```
User speaks into browser microphone
!“
WebAudio API captures audio and streams chunks via WebSocket
!“
webrtcvad on server detects end of speech
!“
Audio sent to Together.ai — openai/whisper-large-v3
!“
Transcription returned: 1.13 seconds
!“
Text sent to Alfred AI with full context and all 87 tools available
!“
Alfred processes, executes tools if needed, generates text response
!“
Text sent to Together.ai — hexgrad/Kokoro-82M
!“
Audio returned: 0.73 seconds
!“
Audio streamed back to browser via WebSocket
!“
Browser plays Alfred's voice
```

Total round-trip: approximately 2 seconds from end of user speech to start of Alfred's voice.

Server-Side Components

Voice WebSocket Server — Flask-SocketIO application. Manages audio streaming in, VAD detection, STT call to Together.ai, response generation via Alfred agent, TTS call to Together.ai, audio streaming out.

Voice API Handler — Together.ai client for Whisper transcription and Kokoro synthesis. Handles streaming audio chunks and reassembly.

Alfred Agent Bridge — Connects voice transcription to the existing Alfred agent. All 87 tools remain available during voice conversation. Alfred can deploy DNS, fix code, create databases, scan for malware — all while speaking.

Browser-Side Components

Audio Capture — WebAudio API. Standard browser API, no library needed. Captures microphone input and streams chunks to server.

WebSocket Client — Sends audio to server, receives audio response, handles connection management.

Audio Playback — Plays Alfred's voice response as it arrives. Streaming playback starts before the full audio is received.

Voice UI — Waveform visualisation during listening and speaking. Push-to-talk toggle. Voice model selector. Visual indicator of Alfred processing.

Speed Reference — All Tested Live

Component	Model	Tested Speed
Speech to text	openai/whisper-large-v3	**1.13 seconds** for 12.6 seconds of audio
Live chat voice	hexgrad/Kokoro-82M	**0.73 seconds**
Quality voice	cartesia/sonic-3	**3.93 seconds**
Expressive voiceover	canopylabs/orpheus-3b-0.1-ft	**8.8 seconds**

Voice Model Strategy

Live conversation: Kokoro-82M — 0.73 seconds. Natural conversational speed. Default for all live chat.

Longer responses: Cartesia Sonic 3 — richer, fuller voice. Used when Alfred is giving a detailed explanation or walking through a complex task.

Voiceover generation: Orpheus 3B — most expressive and emotional. Used for generate_audio tool output, not live conversation.

Voice cloning: XTTS v2 on server — 6-second sample clones any voice. Used for agency white label and customer site voiceovers in a specific person's voice.

What Voice Unlocks

Feature	Detail
Talk to Alfred live	Browser mic, 2-second round-trip, natural conversation
Alfred speaks every reply	All responses optionally voiced
Voice-controlled hosting	"Alfred, my site is down" — Alfred investigates, fixes, and talks you through it
Clone a custom voice	6-second sample via XTTS v2 — any voice replicated
Customer site voiceovers	"Read my homepage copy as a voiceover" — MP3 generated and delivered
Multilingual	Whisper detects language automatically — 17 languages supported
Agency white label	Each agency's Alfred has a unique cloned voice

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Competitive Comparison

Alfred vs Everyone

Capability	Alfred Now	Alfred After Build	Cursor	ChatGPT	Gemini
Live voice conversation	No	**Yes**	No	Yes	Yes
Acts on voice commands	No	**Yes**	No	No	No
Deploy code live	Yes	Yes	No	No	No
Manage DNS and SSL	Yes	Yes	No	No	No
Create and manage databases	Yes	Yes	No	No	No
WordPress management	Yes	Yes	No	No	No
Malware scanning	Yes	Yes	No	No	No
Semantic codebase search	No	**Yes**	Yes	No	No
Vision — screenshot to code	No	**Yes**	Yes	Yes	Yes
Vision AND deploy fix live	No	**Yes**	No	No	No
Generate video	No	**Yes**	No	Limited	Limited
Generate voiceovers	No	**Yes**	No	No	No
Clone a voice	No	**Yes**	No	No	No
Edit images with text	No	**Yes**	No	Partial	Partial
Download from 1,864 sites	Yes	Yes	No	No	No
Agentic auto-loop	Partial	**Yes**	Yes	Partial	Partial
Manage billing and invoices	Yes	Yes	No	No	No
Execute SQL from chat	No	**Yes**	No	No	No
Process and edit video	No	**Yes**	No	No	No
Fetch URL as context	No	**Yes**	Yes	No	No
French language support	Partial	**Full**	No	Yes	Yes

What Cursor Still Does Better After This Build

Two things only — both are editor-level UI features, not agent or AI features:

Inline code completion while typing: Ghost text as you type. Requires LSP integration at the editor level. Separate roadmap item, not part of this build.

Unified multi-file diff view: All file changes from a refactor shown as one diff. Currently Alfred shows file by file. Pure IDE UI change, not an AI or backend item.

Everything else — Alfred wins or ties.

Complete Reference

The Four Pip Installs

```
pip3 install websockets flask-socketio webrtcvad beautifulsoup4
```

The One System Install

```
apt-get install espeak-ng
```

Together.ai Endpoint and Key

Base URL: https://api.together.xyz/v1
API Key: Already configured as TOGETHER_API_KEY environment variable on server
Status: Tested live — all models confirmed available

Do Not Reinstall

The following are already on the server and working: PyTorch 2.8.0 with CUDA 12.8, HuggingFace Transformers 4.57.6, ONNX Runtime 1.23.2, Coqui TTS with XTTS v2 model fully downloaded, AudioCraft, FFmpeg 7.0.2, MoviePy 2.1.2, ImageMagick 7.1.1, Pillow 11.3.0, vtracer, Librosa, Basic Pitch, yt-dlp 2026.02.04, Scikit-learn, SciPy, NumPy, Flask, aiohttp, requests, wkhtmltopdf, Node.js 20.20.0, Composer 2.9.5, MariaDB 10.6.25, WP-CLI 2.12.0, PHP 8.2 and 8.3, Git 2.34.1, curl, wget, rsync, OpenSSL.

The Bottom Line

Together.ai is the AI engine. The server is the hands. Four pip installs. One system package. One API key already working.

That is the entire gap between Alfred today and Alfred as the most capable AI agent available anywhere in the world — one that talks to you live, manages your hosting, generates video, clones voices, searches your codebase semantically, reads screenshots, edits images, processes video, and does all of it in 17 languages.

No other product has this combination. Build it and GoCodeMe is not competing with hosting companies anymore. It is in a category of one.